

Go Green Foundation!

Fall 2026

VIP – Urban Sustainability

Hyungkyu (Tom) Lee, Ray Ramadhar

Prof. Baglione & Prof. Rundell

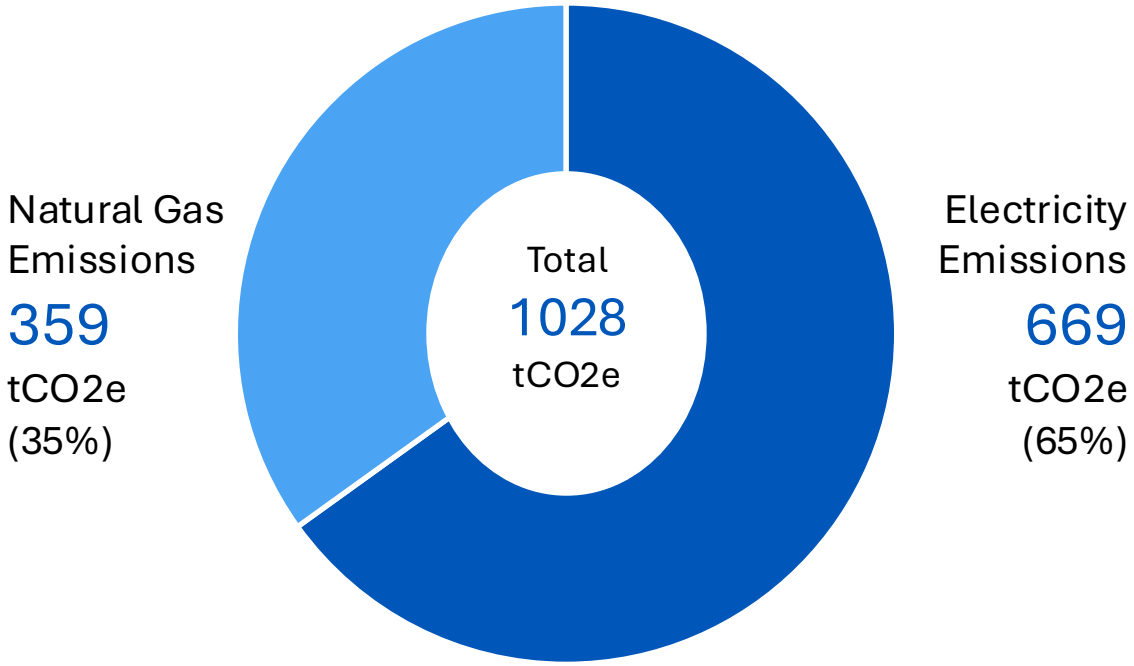
Cooper Union Foundation Building

BULIDING SNAPSHOT



- Built in 1859
- Today designated as a New York City landmark
- Total building area: 190,144 sq ft

EMISSIONS BY SOURCE IN 2025



Foundation Building Emissions

tCO_2e based on LL 97 Carbon CC

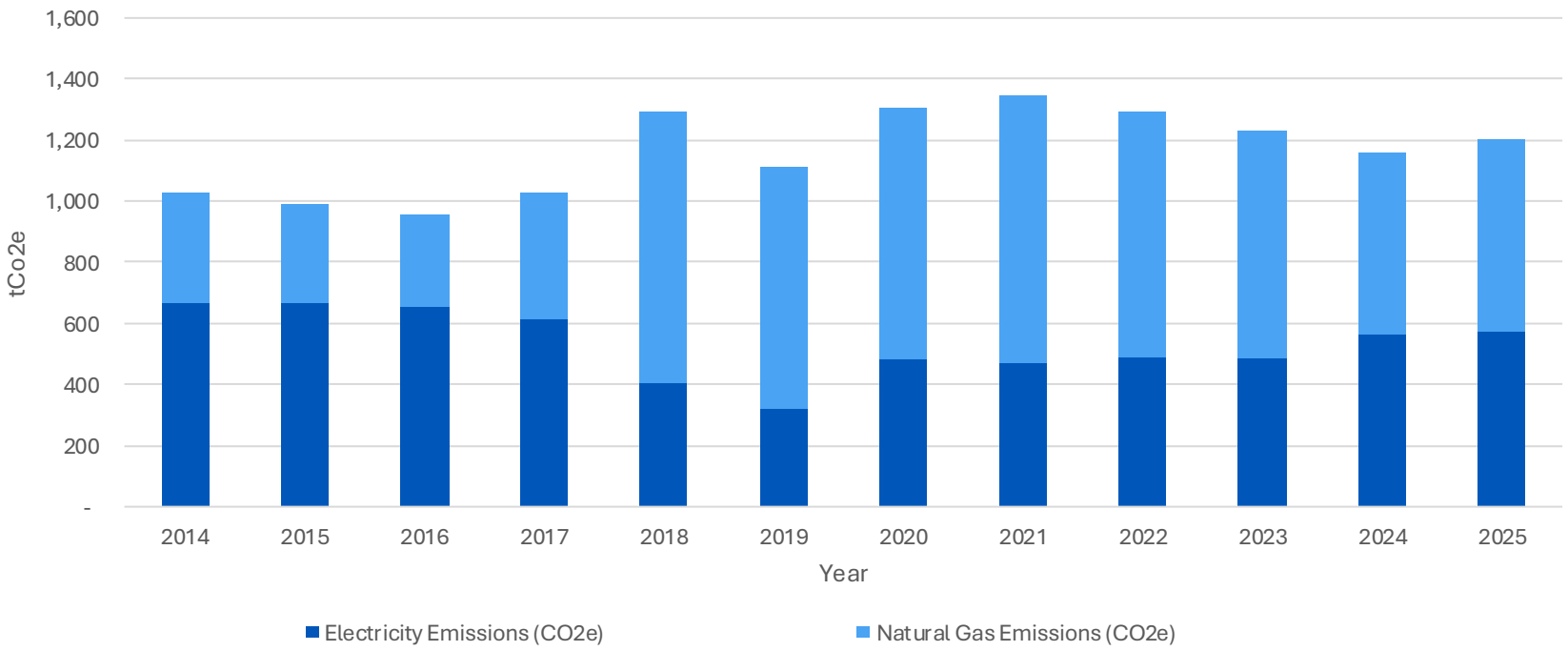
2025 Total Emissions

1,028 tCO_2e

2025 Emissions Intensity (EUI)

11.92 tCO_2e / sq ft

Campus Total Emissions by Source Trends (tCO_2e)



Local Law 97 Goal: Reach

399 tCO_2e

by 2030

235 tCO_2e

by 2035

34 tCO_2e

by 2040

0 tCO_2e

by 2050

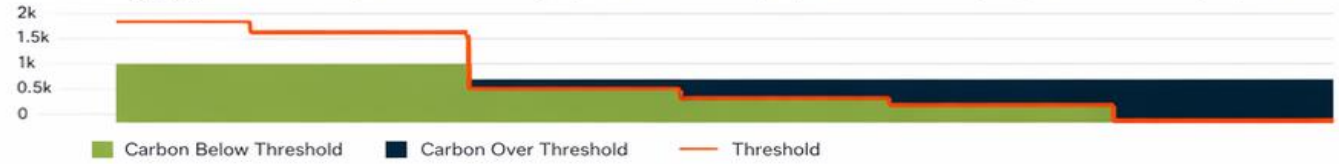
The Foundation Building is predicted to face fines in the 2030s due to Local Law 97 penalties.

1 **Foundation Building**

Old School control and efficiency measures

Annual Carbon Threshold Summary

tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	989	655	655	655	655
Threshold	1,877	399	235	34	0
Est. Penalty ($/yr)	$0	$68,610	$112,608	$166,395	$175,610



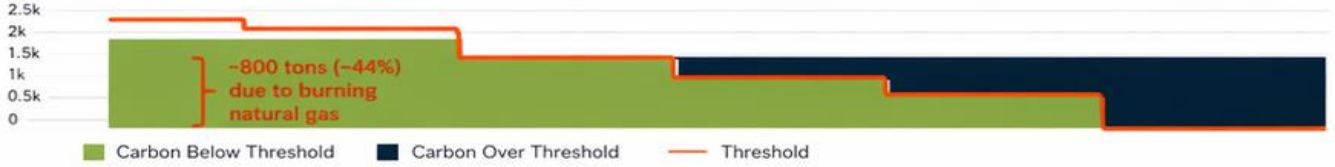
$68,610 / yr
LL97 penalties expected by 2030

2 **41 Cooper Square**

High Natural Gas: Electricity emissions

Annual Carbon Threshold Summary

tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	1,839	1,322	1,322	1,322	1,322
Threshold	2,335	1,387	995	577	0
Est. Penalty ($/yr)	$0	$0	$87,661	$199,674	$354,310

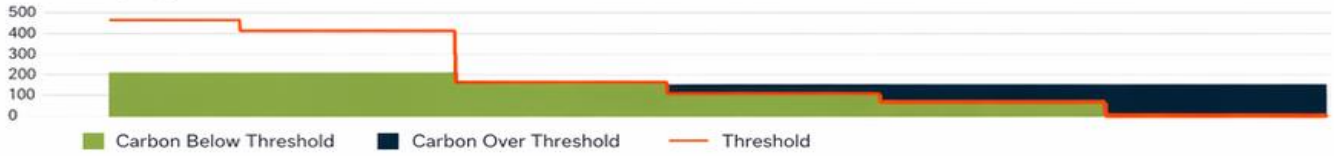


$87,661 / yr
LL97 penalties expected by 2035

3 **Residence Hall**

Annual Carbon Threshold Summary

tCO_2e/yr	2024–2029	2030–2034	2035–2039	2040–2049	2050+
Emissions	247	165	165	165	165
Threshold	465	161	96	46	0
Est. Penalty ($/yr)	$0	$885	$18,531	$31,833	$44,138



$885 / yr
LL97 penalties expected by 2035

What is an Air Handling Unit?

Overview

- An Air Handling Unit (AHU) is a major component of a building's heating ventilation and cooling (HVAC) system.
- The purpose of this component is to maintain comfortable temperatures and clean, fresh indoor air.
- It does this by filtering, heating or cooling, and circulating air throughout the building.

Different Types

- *Singlezone* – Serves a single thermal zone
- *Multizone* – Serves multiple controlled areas
- *Constant Volume* – Constant Airflow
- *Variable Air Volume* – Varies flow to control temp









Newer AHU connected to the BMS

Older AHU not connected to the BMS

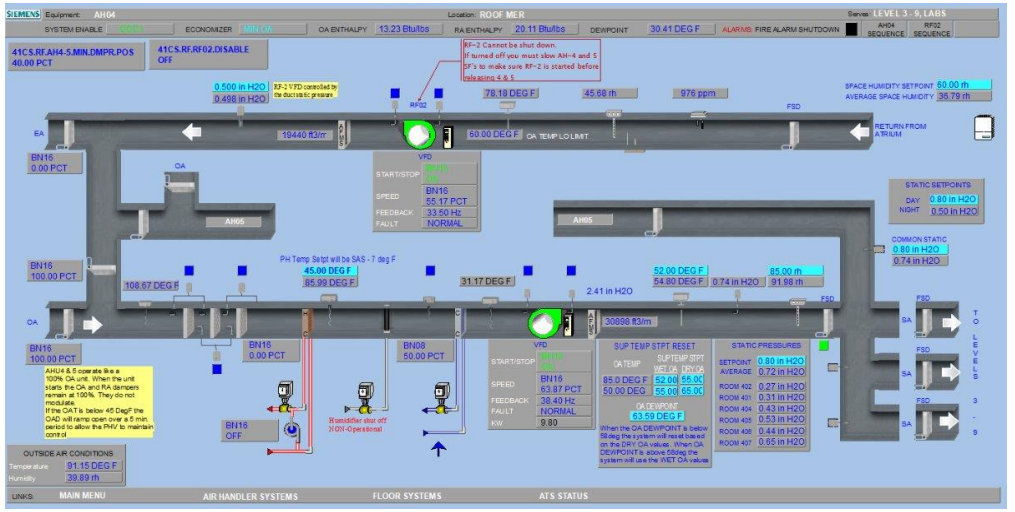
What is Building Management System (BMS)?

Overview

- Computer platform that monitors and controls a building's mechanical and electrical systems. Systems include:
  - HVAC
  - Lighting
  - Power
  - Fire Safety

Specific to AHU's

- All of 41 CS's AHU's are connected to the BMS and are all VAV.
- FB's AHU's are a mix of SZ, MZ, CV, and VAV
- Some use old controls such as pneumatic controls and are not connected to the BMS



BMS screenshot of AHU-4 from 41 CS (VAV)



Pneumatic Control from MZ-3

Proposal: quantify the 63% AHU monitoring blind spot this semester

Install current transformers on selected non-BMS units to convert an unknown load into an actionable energy baseline

CURRENT STATE

19 of 30

AHUs are outside the BMS

PROPOSED PLAN

Use Current Transformers

load monitoring on selected AHUs

DECISION VALUE

Actionable

energy and control recommendations

EACH MONITORED AHU WILL PRODUCE FOUR VALUES

DEMAND

kW

Typical and peak load

CONSUMPTION

kWh

Daily and weekly use

SCHEDULE

Start / stop

Actual operating hours

EXCEPTION

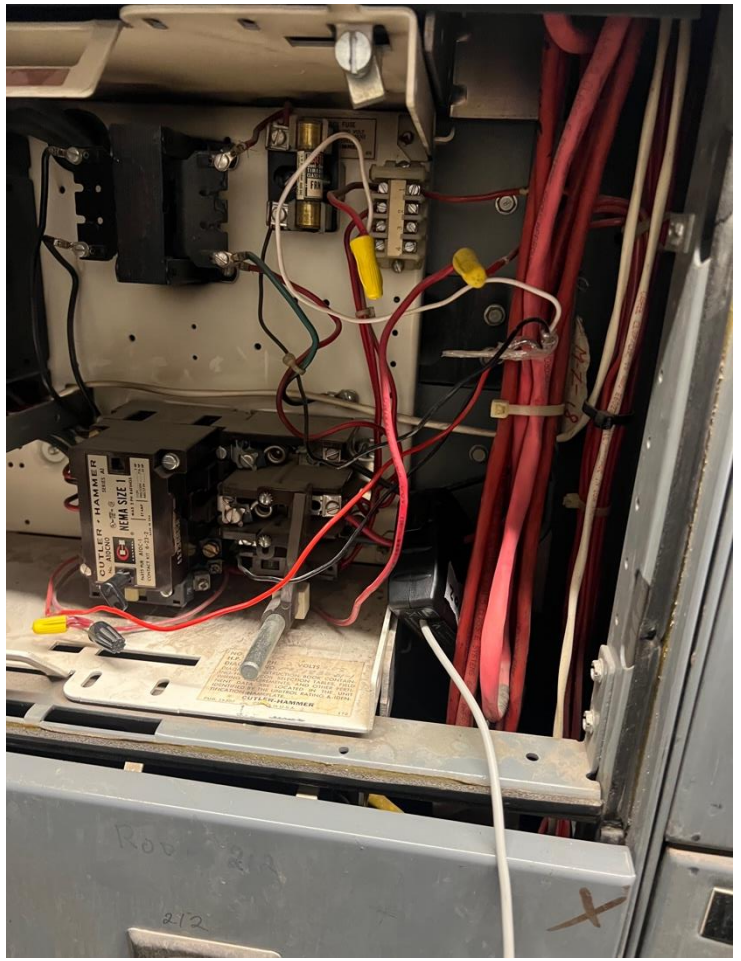
After-hours

High or unexpected load

Current Transformer Monitoring in Action



Current Transformer Sensor and Logger

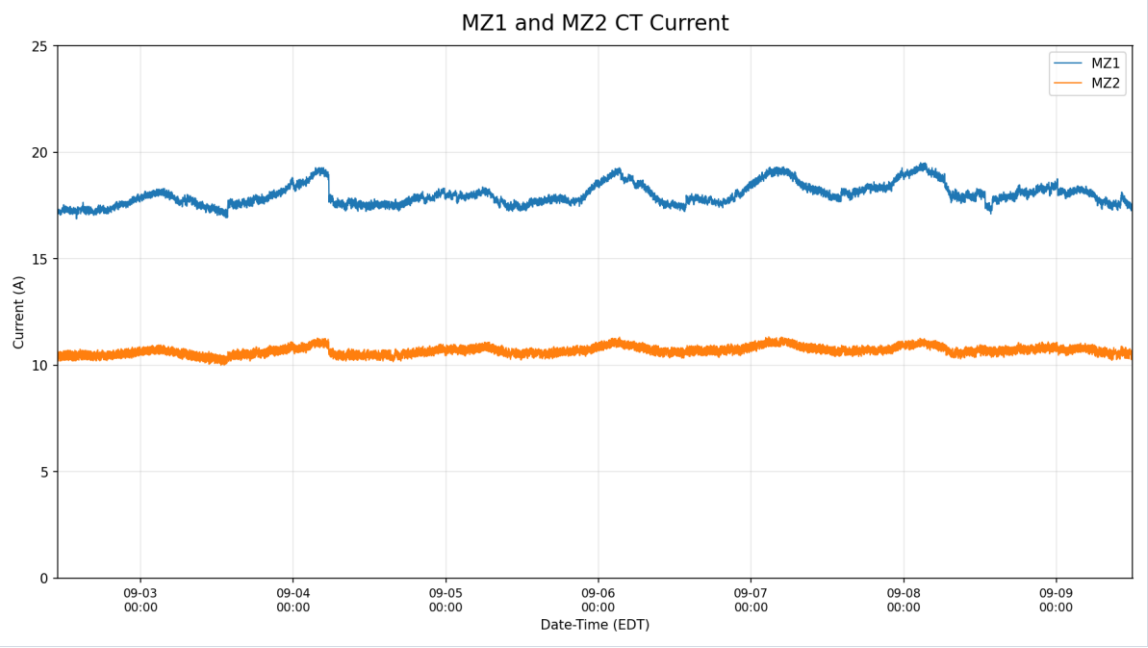


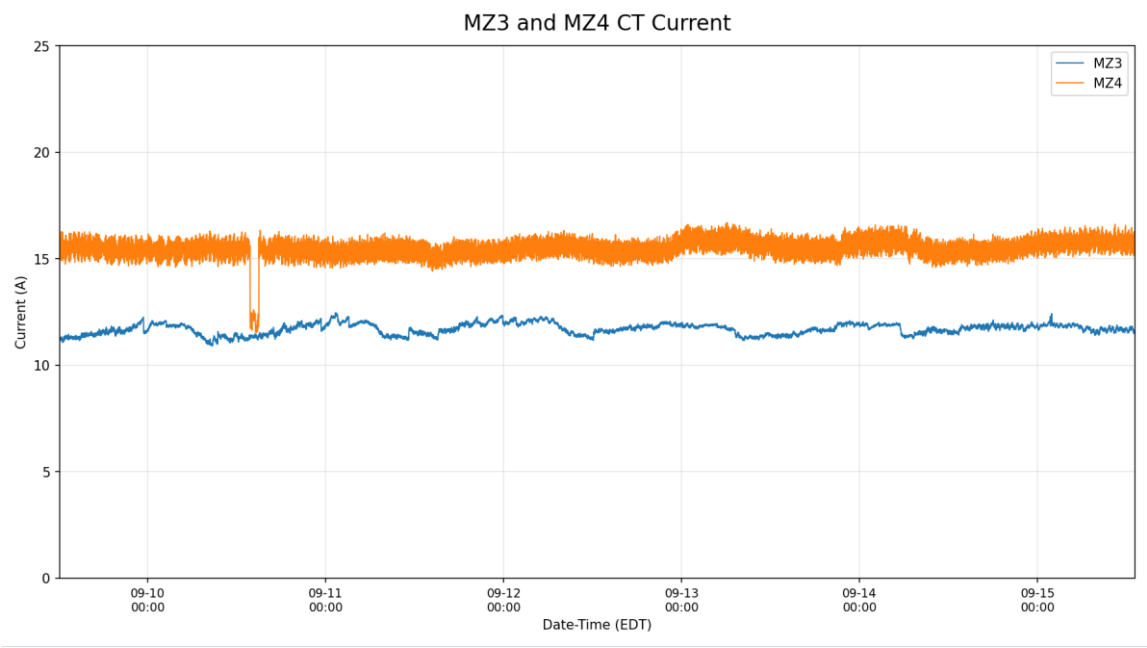
CT installed inside the AHU panel



CT and logger recording current

CT current readings | MZ1–MZ4





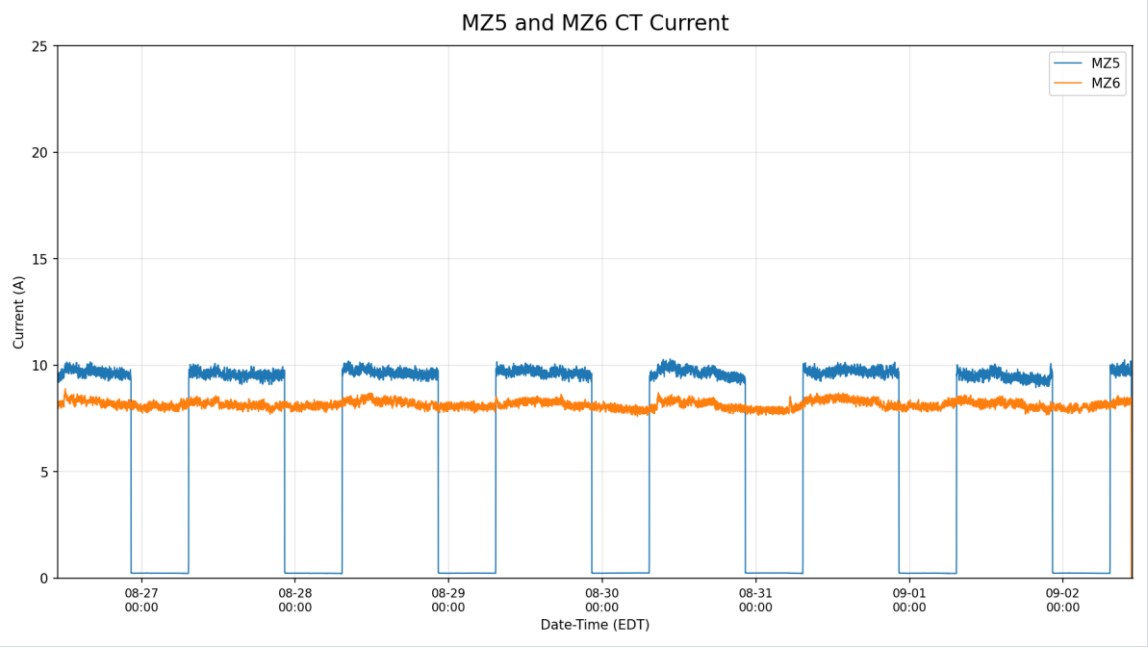
MZ1–MZ2

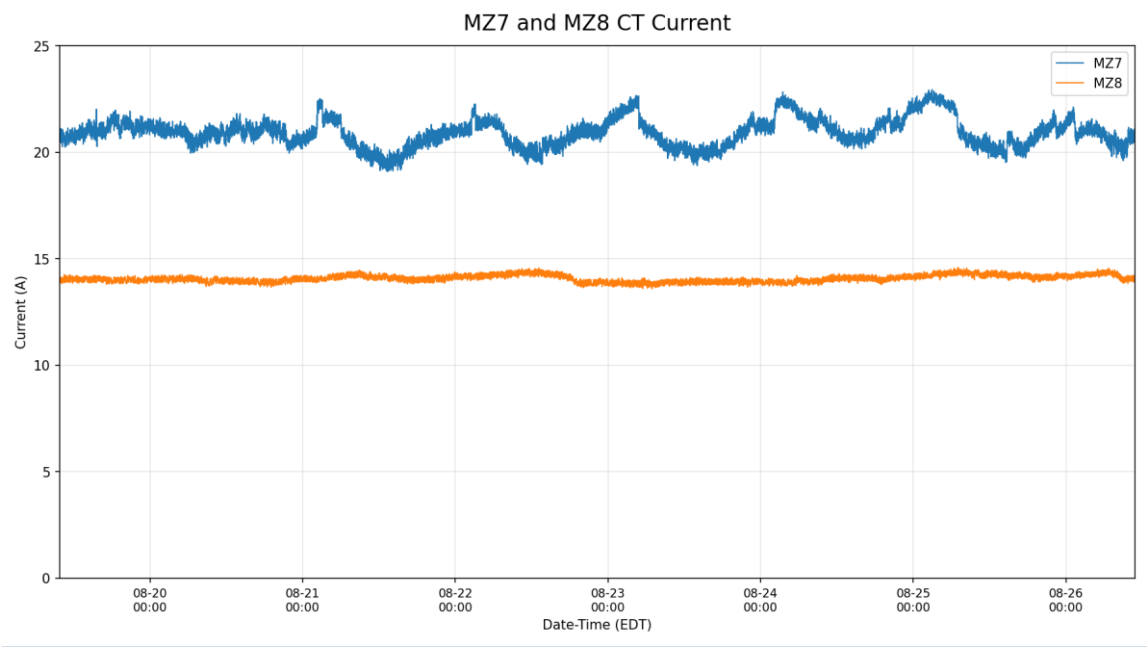
UNIT	SERVED AREA	CURRENT PROFILE
MZ1	7th Floor	17–19 A · variable
MZ2	7th Floor	10–11 A · constant

MZ3–MZ4

UNIT	SERVED AREA	CURRENT PROFILE
MZ3	6th Floor	11–12 A · variable
MZ4	6th Floor	15–16 A · constant

CT current readings | MZ5–MZ8





MZ5–MZ6

UNIT	SERVED AREA	CURRENT PROFILE
MZ5	5th Floor	0-10 A · night setback
MZ6	5th Floor	~8 A · constant

MZ7–MZ8

UNIT	SERVED AREA	CURRENT PROFILE
MZ7	5th Floor	19–22 A · variable
MZ8	5th & 6th Floors	~14 A · constant

Objectives

Objectives	Description	OUTPUT
1 Verified AHU inventory	Operating status, location and BMS connection confirmed for all 30 AHUs	**30-unit inventory**
2 Measured load profiles	CT time series collected for selected units during occupied and unoccupied periods	**kW + run time**
3 Non-BMS AHU energy estimate	Daily/weekly kWh calculated and non-BMS share of HVAC electricity estimated	**kWh + HVAC %**
4 Prioritized fixes	2–3 monitoring, scheduling or control changes tied directly to measured findings	**2–3 actions**

Appendix | Detailed Foundation Building AHU Inventory

Total Number of AHU: 30 traced units

AHUs on BMS: 11

AHU Types: Single Zone, Multizone, Constant Volume, Variable Air Volume

Unit	Unit Audit Report	Location	Service	On BMS	Make	Air Flow	CV/VF
						Test CFM	CAV/VAV
MZ-1	MZ-1	Roof (800)	1) Peter Cooper Suite (800, 801); 2) Exhibition Area, Offices (700, 701, 703 704); 3) Office (702)	No	McQuay	12,612	CV CAV
MZ-2	MZ-2	MER (721)	1) Corridor (720); 2) Rm 712; 3) Rm 713; 4) Rm 714; 5) Rm 715	No	McQuay	2,850	CV CAV
MZ-3	MZ-3	MER (621)	1) Rm 621; 2) Rm 613; 3) Rm 614; 4) Rm 615;	No	McQuay	2,920	CV CAV
MZ-4	MZ-4	MER (632)	1) Rm 629; 2) Rm 630; 3) Rm 631; 4) Rm 632;	No	McQuay	15,354	CV CAV
MZ-5	MZ-5	MER (534)	1) Corridor (521) 2) Corridor (520), Storage (534); 3) Silk Screen Shop NE (544); 4) Silk Scren Shop NW (544); 5) Classroom (537); 6) Classroom (543);	No	McQuay	8,862	CV CAV
MZ-6	MZ-6	MER (534)	1) Classroom (515); 2) Graphiscs Shops (514, 514A); 3) Graphics Shops (512, 513);	No	McQuay	5,958	CV CAV
MZ-7	Mz-7	MER (534A)	1) Photo Studio NE (530); 2) Photo Studio SE (530); 3) Classroom (539); 4) Classroom (541); 5) Projection Booth (538); 6) Projection Booth (540);	No	McQuay	13,005	CV CAV
MZ-8	MZ-8	MER (534)	1) Exhibition Area (600); 2) Office (601); 3) Exhibit Area (500); 4) Exhibit Area (501);	No	McQuay	11,059	CV CAV
SZ-8	SZ-8	Office (213A)	Exhibition Area, Corridor (Rm 230, 220)	No	McQuay	4,201	CV CAV
MA-2	MA-2	F15 MER	Make-up Air for 5th Floor Paint Shop	No	Trane	1,902	CV CAV

AH-4A	AH-4A	Woodwork Shop (430)	VAV-401: Woodwork Shop (430); VAV-402(RHC): Hi Vent Paint / Plaster Rm 9421); VAV-403: 3D Design Studio (415); VAV-404: Sculpture Studio (414); VAV-405: Sculpture Studio (413); VAV-406: Plastic Rm (412)	Yes	Carrier	15,293	VFD VAV
AH-4B	AH-4B	Woodwork Shop (430)	VAV-407: Woodwork Shop (430); VAV-408: Woodwork Shop (430);	Yes	Carrier	2,631	VFD VAV
AH-4C	AH-4C	Woodwork Shop (430)	VAV-409: Sculpture Studio (401A); VAV-410: Sculpture Studio (400);	Yes	Carrier	496	VFD VAV
AH-3A	AH-3F	Office (338)	VAV-309-thru 314: Architecture Studio (330)	Yes	Carrier	5,270	VFD VAV
AH-3B	AH-3B	Office (337)	VAV-315: Office (338); VAV-316: Student Workspace (313); VAV-317: Student Workspace (314); VAV-318: Student Workspace (312);	Yes	Carrier	2,359	VFD VAV
AH-3D	AH-3D	Office (335)	VAV-301-thru 308: Architecture Studio (330)	Yes	Carrier	5,512	VFD VAV
AH-3C	AH-3C	Office (332)	1) Lobby Area (300); 2) Admin Offices (301);	Yes	Carrier	4,569	VFD VAV
AH-3F	AH-3F	Corridor (320)	VAV (w/RH): Lecture Hall (315)	Yes	Carrier	3,870	VFD VAV
MZ-18	MZ-18	MER (212A)	1) Classroom (215); 2) Office (214)	No	Carrier	3,644	CV CAV
MZ-19	MZ-18	MER (212A)	1) Office (212); 2) Office (213)	No	Carrier	3,715	CV CAV
MZ-20		Mezz MER	1) Library Workroom; 2) Mezz Corr; 3) FL2 Corr	No	Carrier	N/A	CV CAV
MZ-21		Mezz MER	1) A/V Center; 2) Seminar Rm; 3) Library Workroom	No	Carrier	N/A	CV CAV
AH-3E		Corridor (320)	VAV (w/RH): Lecture Hall (315)	Yes	Carrier	N/A	VFD VAV
SZ-1	AHU 1	Library Corridor (135)	Main Library (Rm 130)	No	Carrier	1,435	CV CAV
SZ-2	AHU 2	Library Corridor (135)	Library Corridor (Rm 135)	No	Carrier	2,134	CV CAV
SZ-3	AHU 3	Lobby MER (202A)	Lobby/Elevators (Rm 100)	No	Carrier	2,699	CV CAV
SZ-4	SZ-4	Cellar MER (B13)	Stacks (Rm 110-115)	No	McQuay	6,000	CV CAV
SZ-5	SZ-5	Sub-Cellar MER (B18)	Great Hall NE (RM B24)	Yes	McQuay	5,567	CV CAV
SZ-6	SZ-6	Sub-Cellar MER (B18)	Great Hall SW, Lobby, Stairwell (RM B24, B00, B07) (NOT OPERATIONAL)	Yes	McQuay	3,988	CV CAV
SZ-7	SZ-7	Classroom (214A)	Offices, Corridor (RM 237-243)	No	McQuay	2,500	CV CAV
SZ-8	SZ-8	Office (213A)	Exhibition Area, Corridor (Rm 230, 220)	No	McQuay	4,201	CV CAV
MA-2	MA-2	F15 MER	Make-up Air for 5th Floor Paint Shop	No	Trane	1,902	CV CAV

Thank you !